Yenatfanta Shifferaw Bayleyegn

AI Researcher | Data Scientist | Data Engineer
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Professional Summary

AI Researcher, Data Scientist, and Data Engineer with over 10 years of experience designing, engineering, and deploying production-grade AI systems across robotics, healthcare, blockchain, and financial technology. Experienced in scalable data engineering, machine learning, distributed software architectures, and cloud platforms, with a passion for translating cutting-edge technology research into reliable industrial solutions that create measurable real-world impact.

Core Technical Skills

- **Programming:** Python, R, SQL, Bash
- **Artificial Intelligence & Machine Learning:** Deep Learning, NLP, LLMs, Computer Vision, TensorFlow, RASA, AWS SageMaker
- **Data Engineering:** ETL, BigQuery, PostgreSQL, dbt, Trino
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Prometheus, Grafana
- **Visualization:** Power BI, Google Data Studio, QuickSight, R Shiny
- **Software Engineering:** Git, REST APIs, PyTest, GoTest, CI/CD

Work Experience

AI Researcher & Developer | SingularityNET Jan 2025–Present

- Develop LLM- and ITP-based solutions for automated theorem proving and formal verification.
- Build autoformalization pipelines to transform mathematical literature into formal proofs.
- Develop AI/ML components for language technologies.
- Research AGI cognitive architectures and integrate NLP models into AI pipelines.
- Collaborate on production-ready AI systems for next-generation intelligent applications.

Senior Data Scientist | Software Engineer | EthSwitch S.C. Sep 2023–Sep 2025

- Designed and deployed ML-powered credit scoring models using financial, social media and psychometric data.
- Engineered scalable ETL pipelines and data infrastructure for analytics and ML.
- Adapted AI solutions to regulatory and privacy requirements for emerging markets.

Data Scientist | Engineer | Input Output Global Apr 2019–Nov 2024

- Developed blockchain analytics and prediction models using AWS SageMaker and Amazon QuickSight, enabling data-driven insights from decentralized systems.
- Designed interactive R Shiny dashboards integrating machine learning models to support real-time analytics and decision-making.
- Optimized dbt/Trino/BigQuery pipelines and monitoring with Prometheus/Grafana.

SME in ML & AI | Excellerent Technology Solutions Dec 2019–Jan 2024

- Built Azure GPT-based medical assistant.
- Designed and developed NLP resume parser and Android ML asset tracker.
- Provided analytics solutions for COVID-19 and farming.

Senior AI Researcher | Ethiopian Biotechnology Institute Mar 2017–Apr 2019

- Developed CNN-based kidney abnormality detection systems.
- Built deep learning solutions for healthcare and agriculture.
- Led AI training sessions and robotics initiatives.

AI Researcher | Developer | iCog Labs Jun 2014–Jan 2019

- Contributed to the development of **Sophia**, the internationally recognized humanoid robot, by implementing computer vision, sound recognition, and real-time perception systems.
- Applied OpenCog and AtomSpace/MOSES algorithms to intelligent robotic systems.

Education

M.Sc. Computer Engineering | Addis Ababa Institute of Technology 2008–2011
B.Sc. Electrical Engineering | Jimma University 2004–2007

Publications

- The Parent-Child Framework: Biologically Inspired Bias-Free Machine Learning Model
- Limitations of Proof of Stake Algorithm in Blockchain: A Review
- Hybrid Intelligent System for Lung Cancer Type Identification

Certifications & Professional Training

- AWS Certified Cloud Practitioner
- Prompt Engineering
- DevOps Fundamentals
- Kubernetes
- Database Systems
- Leadership Development

Professional Competencies

- Project Management & Agile Delivery
- Cross-functional Collaboration & Stakeholder Communication
- Mentorship, Coaching & Technical Knowledge Sharing
- Collaboration Tools: Jira, Confluence, Microsoft Teams, Slack

Leadership & Professional Activities

- Ambassador: Women in Data Science (WiDS) Ethiopia
- Poster Judge: Deep Learning Indaba
- Team Advisor: Women-focused AI Hackathon
- Recipient of competitive research visit scholarships and conference participation grants from CVPR, ICCV, WiML, Black in AI, NeurIPS, and Deep Learning Indaba.

Community Engagement

- Volunteer supporting vulnerable mothers and children through providing food and clothes
- Volunteer tutoring girls in STEM
- Volunteer reading companion for elderly community members